

D.I.R.T.T

Digital Instrument for Remote Terrain Testing

An affordable, autonomous robot that collects and analyzes soil samples across crop fields.

Team Members



Nolan Brechtel

Project Manager | Navigation Lead



George Utz

Chief Scientist | Power & Charging Lead



Cara Belew

Data Backup & Control Lead



Zachary Morrow

Soil Testing Lead



Rebecca Ueltschey

Drivetrain Lead



Dr. Christopher Hudson

Faculty Advisor

The Problem

25,000 people die every day from hunger and related causes [1]

$41.5B projected U.S. agricultural trade deficit, FY2026 [2]

49% of U.S. crop fields show poor health or nutrient imbalance [3]

Who We Help

Agricultural Research Labs
Need affordable, autonomous sampling to reduce technician hours in the field

Independent Farms
Need reliable soil data without the high cost of existing robots

Field Technician
Need to reduce repetitive, manual sampling work

Competitive Advantage

How soil gets tested today

Manual sampling
A technician walks the field with a hand probe, and samples are mailed to a lab. Results never reach the farmer in the field.

Commercial sampling robots
Rogo's SmartCore is a converted Bobcat skid-steer with RTK GPS and LiDAR, sold as a service at $3.50–4.50 per acre [4].

The gap
Accurate, but every survey is a recurring cost. Small farms and labs sample less often than they should.

What D.I.R.T.T does differently

About $1,000 to build
Owned by the farm or lab, with no per-acre service fee.

Answers in the field
pH, EC, temperature and moisture sent back live over LoRa radio.

Runs on sunlight
A 30 W solar panel stretches each survey day.

Small and light
Under 20 kg, so it fits in a pickup bed and stays light on the soil.

Trade-off: a screening tool, not lab-grade. It points farmers to the spots worth a lab test.

System Diagram

ON THE ROBOT

Navigation
Nolan Brechtel

- SpeedyBee F405 Wing running ArduPilot Rover
- u-blox M10 GPS + compass on a mast
- Geofence, e-stop, bump switches, RC override

MAVLink UART

Data & Control
Cara Belew

- Raspberry Pi 3 A+ runs the mission
- LoRa Bonnet with OLED status screen
- Heartbeat, retries, return-home logic

915 MHz LoRa link

test points · results · heartbeat

BASE STATION

Base Station
Data & Control · Cara Belew

- Raspberry Pi 5 with a LoRa Bonnet
- Hands out one test point at a time
- Watches the robot's heartbeat; sends COME HOME
- Laptop shows a live map and saves results to CSV

PWM to motor driver

RS485 · I2C · GPIO

Drivetrain
Rebecca Ueltschey

- 4× 12 V gearmotors with encoders
- Dual-channel motor driver, skid steer
- 5052 aluminum chassis

Soil Testing
Zachary Morrow

- L16 linear actuator, 200 N, position feedback
- 4-in-1 probe: pH, EC, temp, moisture
- Actuator stall = rock strike

12 V
5 V

Power & Charging
George Utz · feeds every subsystem

- 30 W solar panel → PWM charge controller
- 12.8 V LiFePO4 battery, fused branches
- 12 V → 5 V regulator; plug-in charge port

Features

F1 Drives itself

Navigates to GPS test points across a field with no driver.

F2 Tests the soil

Pushes a probe into the ground and stops if it hits a rock.

F3 Four readings per point

pH, electrical conductivity (EC), temperature and moisture.

F4 Sun-powered

A solar panel tops up the battery; plug in to charge between runs.

F5 Comes home on its own

If the radio link drops, it pulls the probe and drives back.

F6 A field heat map for the farmer

The base logs each reading with its location for a map view and file export.

Constraints

C1

≤ $1,000

Total project budget

C2

Apr 2027

Field-ready prototype

C3

Communication

Data must be transmitted wirelessly

C4

Weather Proof

The enclosure must protect the electronics from dust particles larger than 2.5mm and water spraying from a 60-degree angle

C5

Solar Charging

No power out in the field.

Requirements

ID	Requirement	Target (measurable)	Feature	How we verify it
R1	Reach each commanded test point	Within 5 m of the target	F1	GPS log vs. a surveyed marker
R2	Drive across the field	≥ 0.5 m/s cruise on grass	F1	Timed 20 m run
R3	Climb slopes at full load	15% grade at 20 kg	F1	Ramp test
R4	Stop on e-stop	Motors stopped in ≤ 3 s	F1, F2	Timed video, 10 trials
R5	Stay inside the field boundary	0 geofence breaches in 10 runs	F1	Simulator + field runs
R6	Insert, read and retract the probe	Full cycle ≤ 60 s	F2, F3	Soil-bin test, stopwatch
R7	Detect a rock strike	Push stops ≤ 3 s after a stall	F2	Buried-brick test
R8	Report four soil values	pH ±0.5, temperature ±1 °C vs. reference	F3	3 soil samples vs. handheld meter
R9	Deliver results to the base	≥ 90% of messages at ≥ 500 m line of sight	F6	100-message range test
R10	Return home when the link drops	Loss detected ≤ 20 s; probe up before moving	F6	Base radio switched off mid-test
R11	Survey on one charge	≥ 1 h of operation with no sun	F4	Battery run-down test
R12	Log and map results	90% of received results saved with GPS + time	F6	Robot log vs. base CSV

Standards

FCC 47 CFR Part 15.247
Unlicensed 902–928 MHz radios: our LoRa link stays in band and at ≤ +20 dBm.

IEC 60529 (IP Code)
Ingress protection: IP33 target for the electronics bay; the soil probe is IP68.

ANSI/A3 R15.08-1-2020
Mobile-robot safety: e-stop, protective stop and manual override.

IEC 62133-2
Lithium battery safety: we buy only LiFePO4 packs certified to it.

MAVLink 2
Message protocol between the Raspberry Pi and the ArduPilot flight controller.

Modbus RTU over TIA-485-A
Serial bus and protocol used to read the 4-in-1 soil sensor.

ISO 10390:2021
Reference method for soil pH, used to check our pH readings.

IPC J-STD-001
Workmanship standard for the soldered connections on the robot.

Component 1: Flight Controller

What it must do

Run ArduPilot Rover: fuse GPS, compass and IMU, drive both skid-steer motor channels, and keep its own failsafes running even if the Pi crashes.

Selection metrics

- Cost ≤ $75, to leave budget for solar
- Runs ArduPilot Rover (tested navigation and failsafes)
- Serial ports for GPS, the Pi (MAVLink)
- Battery voltage and current sensing for low-battery return
- In stock now

	Pixhawk 6C Mini	Matek H743-Slim V4	SpeedyBee F405 Wing
Price	$176.99	$128.99	$67.26
Processor · flash	STM32H743 · 2 MB	STM32H743 · 2 MB	STM32F405 · 1 MB
ArduPilot firmware	Standard build	Standard build	Custom build (trimmed)
Wiring	Plug-in JST-GH	Solder pads	Pads + power board
Battery sensing	Extra power module	Onboard	Onboard, 90 A sensor
Meets ≤ $75	✗	✗	✓

Choice: SpeedyBee F405 Wing. It runs the same ArduPilot Rover software for $80–130 less than the alternatives.

Trade-offs: 1 MB of flash means a custom firmware build (Rover, MAVLink, geofence, failsafes), and the wiring is soldered.

Prices: Pyrodrone, team design summary, RaceDayQuads (Oct 2026) [8]

Component 2: Robot Computer

What it must do

Run the mission on the robot: talk to the base over LoRa, command the flight controller over MAVLink, drive the probe and read the soil sensor.

Selection metrics

- Price ≤ $30
- In stock now
- Low idle power for a solar robot
- Runs Raspberry Pi OS, so the LoRa Bonnet and its Python libraries work
- USB port for the RS485 sensor adapter

	Pi Zero 2 W	Pi 3 Model A+	Pi 5 (1 GB)
Price	$15 (list)	$25	$45
Availability	Sold out	In stock	In stock
Idle power	0.4 W	~1–2 W (est.)	~3 W
CPU · RAM	4× A53 1.0 GHz · 512 MB	4× A53 1.4 GHz · 512 MB	4× A76 2.4 GHz · 1 GB
Works with Bonnet	✓	✓	✓
USB for RS485	Needs OTG adapter	Full-size USB-A	4× USB-A

Choice: Raspberry Pi 3 Model A+ on the robot. Same chip family and software as the Zero 2 W, in stock today, and its USB-A port takes the RS485 adapter directly.

Impact: about 1 W more in the power budget than the Zero 2 W. The LoRa Bonnet sits on a stacking header, so the serial, I2C and GPIO pins stay free. Also considered: Orange Pi Zero 2W (no Raspberry Pi OS).

Component 3: Probe Actuator

What it must do

Push the 4-in-1 soil probe into the ground, hold it while the Pi reads, then pull it out, and tell the Pi when it hits something it can't push through.

Selection metrics

- Push force 100–250 N
- Insert-and-retract cycle ≤ 60 s
- Built-in position feedback
- Runs from 12 V at low current
- Cost ≤ $125

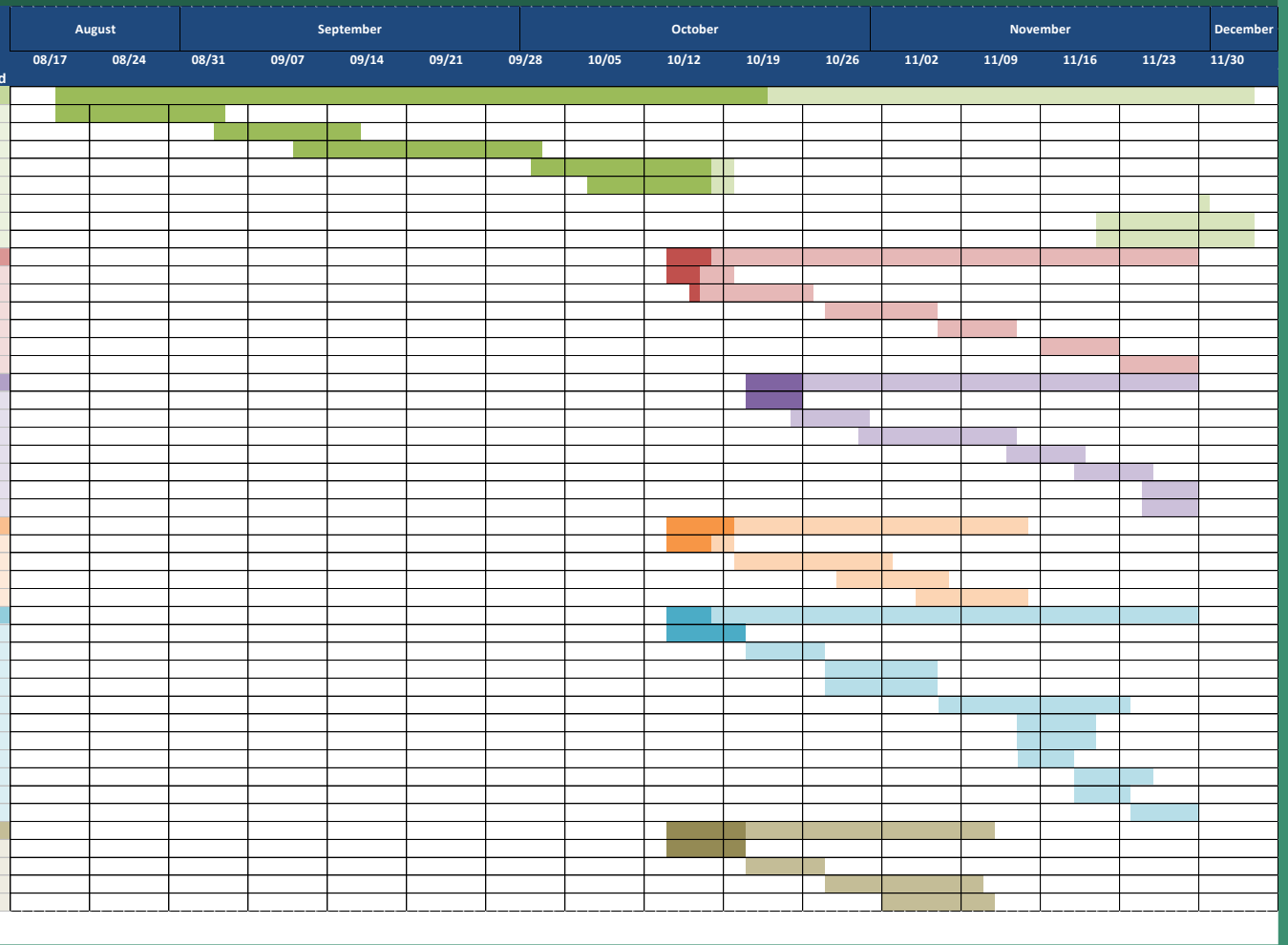
	Actuonix L16 (150:1)	Actuonix P16 (256:1)	Stepper + lead screw
Price	~$100	$106.59	~$172
Push force	200 N (250 N static)	300 N (500 N static)	~250 N
Speed	8 mm/s	2.5–4.8 mm/s	Set by driver
Position feedback	Potentiometer	Potentiometer	None; homing switch
Stall current @ 12 V	0.65 A	1.0 A	Draws holding current
Sealing	IP54	IP54	Unsealed

140 mm stroke takes ~18 s each way, against 31 s or more for the P16) and the lowest current.

Impact: its position feedback lets the Pi detect a stall (a rock), so the load cell and its amplifier were cut. It stands ~8 in above the deck, so the solar panel mount must clear it.

Gantt Chart: Capstone I

D.I.R.T.T. - MSDR Timeline Capstone I							
Milestone description	Assigned to	Support	Progress	Start	Days	End	Parts Ordered
Project management	**All**	**All**	**60%**	**Aug 21, 2026**	**105**	**Dec 4, 2026**	**N/A**
Project Proposal			100%	Aug 21, 2026	15	Sep 4, 2026	
Project Design Requirements (PDR)			100%	Sep 4, 2026	13	Sep 16, 2026	
Problem Statement			100%	Sep 11, 2026	22	Oct 2, 2026	
Design Requirements Specification (DRS)			90%	Oct 2, 2026	18	Oct 19, 2026	
Mid Semester Design Review (MSDR)			90%	Oct 7, 2026	13	Oct 19, 2026	
End-of-semester demo			**0%**	Nov 30, 2026	1	Nov 30, 2026	
Final Design Document			**0%**	Nov 21, 2026	14	Dec 4, 2026	
Final Design Review Presentation (FDR)			**0%**	Nov 21, 2026	14	Dec 4, 2026	
Drivetrain/Chassis	**Rebecca**	**Nolan**	**10.00%**	**Oct 14, 2026**	**46**	**Nov 29, 2026**	**No**
Motor and drive selection			50%	Oct 14, 2026	6	Oct 19, 2026	
Chassis CAD and ground clearance check			10%	Oct 16, 2026	11	Oct 26, 2026	
Fabricate chassis parts as they arrive			**0%**	Oct 28, 2026	10	Nov 6, 2026	
Assemble drivetrain and chassis			**0%**	Nov 7, 2026	7	Nov 13, 2026	
Slope and clearance test			**0%**	Nov 16, 2026	7	Nov 22, 2026	
Mass and clearance verification, fixes			**0%**	Nov 23, 2026	7	Nov 29, 2026	
Control and Data Backup	**Cara**	**George**	**14.29%**	**Oct 21, 2026**	**39**	**Nov 29, 2026**	**No**
Select microcontroller and radio			100%	Oct 21, 2026	5	Oct 25, 2026	
Firmware design: logging and message confirmation			**0%**	Oct 25, 2026	7	Oct 31, 2026	
Base station map and flagging software			**0%**	Oct 31, 2026	14	Nov 13, 2026	
Bench-build radio link			**0%**	Nov 13, 2026	7	Nov 19, 2026	
Onboard logging and message confirmation on hardware			**0%**	Nov 19, 2026	7	Nov 25, 2026	
Radio range test			**0%**	Nov 25, 2026	5	Nov 29, 2026	
Link-loss failsafe test			**0%**	Nov 25, 2026	5	Nov 29, 2026	
Navigation	**Nolan**	**Rebecca**	**20%**	**Oct 14, 2026**	**31**	**Nov 14, 2026**	**No**
GPS and Controller module selection			80%	Oct 14, 2026	6	Oct 19, 2026	
Test-point grid generator			**0%**	Oct 20, 2026	14	Nov 2, 2026	
GPS accuracy test			**0%**	Oct 29, 2026	10	Nov 7, 2026	
Obstacle sensor bench test			**0%**	Nov 5, 2026	10	Nov 14, 2026	
Soil Testing	**Zac**	**Cara**	**9%**	**Oct 14, 2026**	**46**	**Nov 29, 2026**	**No**
Sensor selection			100%	Oct 14, 2026	7	Oct 20, 2026	
Probe insertion concept and force calc			**0%**	Oct 21, 2026	7	Oct 27, 2026	
Bench test sensors			**0%**	Oct 28, 2026	10	Nov 6, 2026	
Build probe insertion mechanism			**0%**	Oct 28, 2026	10	Nov 6, 2026	
Sensor calibration against reference meter			**0%**	Nov 7, 2026	17	Nov 23, 2026	
Insertion force test in soil			**0%**	Nov 14, 2026	7	Nov 20, 2026	
EC normalization to 25 C			**0%**	Nov 14, 2026	7	Nov 20, 2026	
Mount probe on chassis			**0%**	Nov 14, 2026	5	Nov 18, 2026	
Obstruction detection and retract test			**0%**	Nov 19, 2026	7	Nov 25, 2026	
Measurement cycle timing test			**0%**	Nov 19, 2026	5	Nov 23, 2026	
Accuracy verification vs reference meter			**0%**	Nov 24, 2026	6	Nov 29, 2026	
Power/Charging	**George**	**Zac**	**25%**	**Oct 14, 2026**	**28**	**Nov 11, 2026**	**No**
Energy budget			100%	Oct 14, 2026	7	Oct 20, 2026	
Battery and solar panel sizing			**0%**	Oct 21, 2026	7	Oct 27, 2026	
Charge controller			**0%**	Oct 28, 2026	14	Nov 10, 2026	
Interim battery and power distribution for testing			**0%**	Nov 2, 2026	10	Nov 11, 2026	



Capstone I Deliverables

Subsystem / lead	Requirements	Demonstration setup and test method	% Grade	% Earned
Data & Control	(1) Robot Pi and base exchange TEST, OK and RESULT messages over LoRa with ≥ 90% delivery (after retries) at ≥ 500 m line of sight. [8%] (2) Heartbeat detects a dead link within 20 s (3 missed beats of 5 s). [6%] (3) Base logs 100% of received results to CSV with time and GPS position. [6%]	(1) Two Pis with LoRa Bonnets on a campus field 500 m apart send 100 scripted messages; the delivery rate is counted from both logs (data report). (2) Live: the base radio is switched off; the robot's OLED shows LINK LOST and the elapsed time is read from the log. (3) The base CSV is opened on the laptop and checked line by line against the robot's send log.	**20%**	
Soil Testing	(1) Pi reads pH, EC, temperature and moisture over Modbus at least every 5 s. [6%] (2) Readings agree with a handheld reference meter: pH within ±0.5 and temperature within ±1 °C on 3 soil samples. [8%] (3) Actuator pushes the probe into a soil bin and stops within 3 s of stalling on a buried obstruction. [6%]	(1) Live console shows all four values updating from the sensor in a soil sample. (2) Three prepared samples (dry, moist, limed) are measured by our sensor and by the reference meter; a table of differences is provided. (3) Actuator on a 3D-printed test stand over a soil bin; a brick buried 4 cm down triggers the stall stop, timed from the log.	**20%**	
Drivetrain	(1) Chassis fabricated and assembled with all four motors and wheels; mass with ballast ≤ 20 kg. [6%] (2) Under RC control, drives ≥ 0.5 m/s on grass and turns in place. [7%] (3) Climbs a 20% grade with 20 kg on board; ground clearance ≥ 3 in. [7%]	(1) Robot is weighed on a scale with ballast in place. (2) Timed 20 m run on campus grass (3 trials, average speed reported) and a turn-in-place demonstration. (3) Plywood ramp set to a 20% (11.3°) grade with ballast aboard; clearance measured with a tape at the lowest point.	**20%**	
Demo	All 3 subsystems are ready to demo at the time scheduled. Each requirement and corresponding demonstration setups written above have been attempted.		**40%**	
Total			**100%**	

Future Work

Capstone I • Fall 2026

- Prove 3 subsystem on the bench
- LoRa link, soil readings, flight controller in the simulator
- Chassis built and driving
- Solar charging on the bench

Capstone II • Spring 2027

- Integrate everything into one robot
- Autonomous multi-point surveys
- Link-loss, e-stop and geofence tests
- Field trials and final demo

Beyond

- Automatic docking and charging station
- Outrigger legs if lift tests need them
- Nutrient (NPK) sensing and a field heat map
- Several robots per field

Broader Impact

Farmers and labs

A ~$1,000 robot puts field-scale soil data within reach of small farms and research labs that can't pay per-acre sampling services.

Environment

Knowing where soil needs help lets farmers apply fertilizer and lime only where it's needed, cutting waste and nutrient runoff.

Food security

Better-managed soil supports better yields, part of feeding a world where hunger still kills thousands every day [1].

Workforce

It automates the repetitive walk-and-probe work, so technicians spend their time on analysis and decisions.

References

1 World Food Programme (via Xinhua), “WFP says hunger kills more than AIDS, malaria, tuberculosis combined,” ReliefWeb, Jun. 4, 2009.

2 R. Hanrahan, “USDA projects ag trade deficit will fall to $41.5 billion in 2026,” Farm Policy News, Univ. of Illinois, Aug. 29, 2025.

3 Source for the 49% crop-field figure: team to add before presenting.

4 “Robotic soil sampler is slow but consistent” (Rogo Ag SmartCore), The Western Producer, Jan. 2020.

5 PiShop.us and Newark, Raspberry Pi 3 Model A+ product pages, accessed Oct. 2026.

6 raspberry.tips, “Raspberry Pi Zero 2 W sold out: why, for how long, and which alternative,” Sep. 14, 2026; power consumption data, 2026.

7 M. Larabel, “Raspberry Pi 5 2GB price increases to $77.50 USD,” Phoronix, Oct. 1, 2026.

8 RaceDayQuads, SpeedyBee F405 WING APP; Pyrodrone, Holybro Pixhawk 6C Mini; ArduPilot, “Firmware limitations on autopilot hardware”; accessed Oct. 2026.

9 Actuonix L16-140-150-12-P and P16-150-256-12-P specifications, ThingBits and RobotShop, accessed Oct. 2026.



Questions?

Digital Instrument for Remote Terrain Testing

Mid-Semester Design Review
Fall 2026